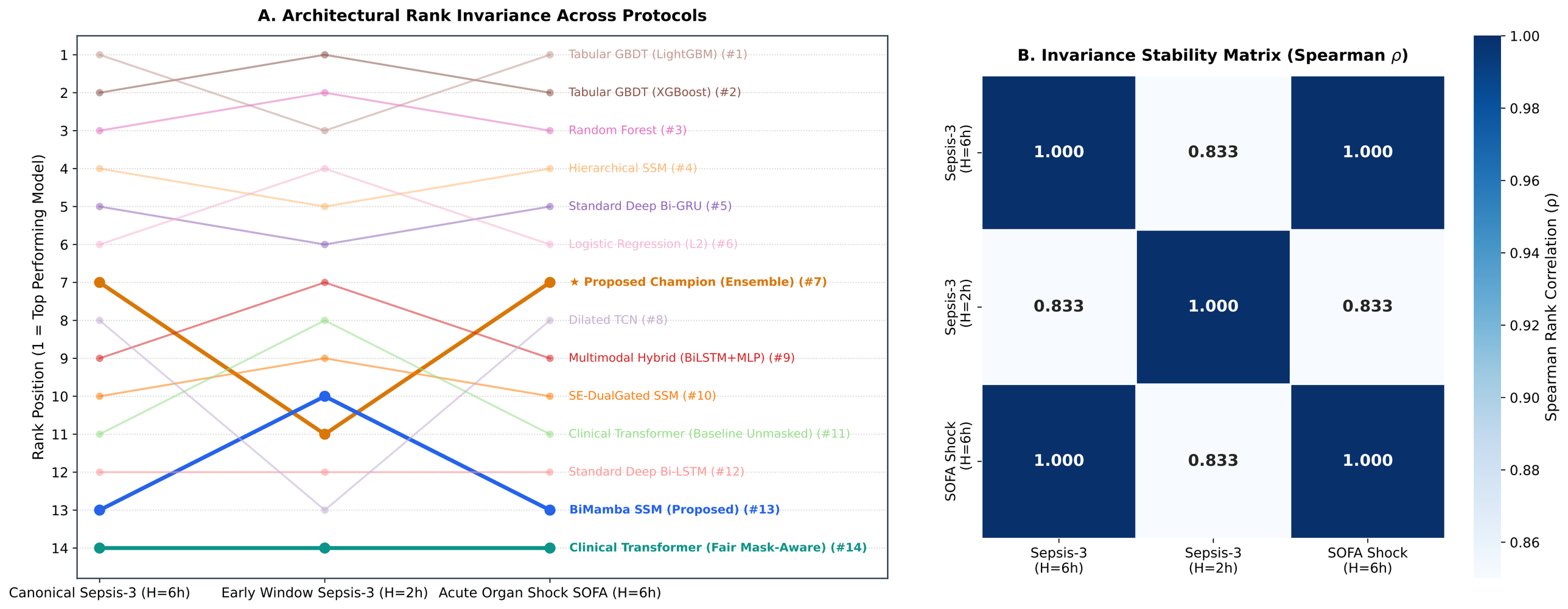
A. Architectural Rank Invariance Across Protocols
Rank Position (1 = Top Performing Model)
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Tabular GBDT (LightGBM) (#1)
Tabular GBDT (XGBoost) (#2)
Random Forest (#3)
Hierarchical SSM (#4)
Standard Deep Bi-GRU (#5)
Logistic Regression (L2) (#6)
★ Proposed Champion (Ensemble) (#7)
Dilated TCN (#8)
Multimodal Hybrid (BiLSTM+MLP) (#9)
SE-DualGated SSM (#10)
Clinical Transformer (Baseline Unmasked) (#11)
Standard Deep Bi-LSTM (#12)
BiMamba SSM (Proposed) (#13)
Clinical Transformer (Fair Mask-Aware) (#14)
Canonical Sepsis-3 (H=6h)
Early Window Sepsis-3 (H=2h)
Acute Organ Shock SOFA (H=6h)
B. Invariance Stability Matrix (Spearman ρ)
Sepsis-3 (H=6h)
Sepsis-3 (H=2h)
SOFA Shock (H=6h)
1.000
0.833
1.000
0.833
1.000
0.833
1.000
0.833
1.000
Sepsis-3 (H=6h)
Sepsis-3 (H=2h)
SOFA Shock (H=6h)
Spearman Rank Correlation (ρ)
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86